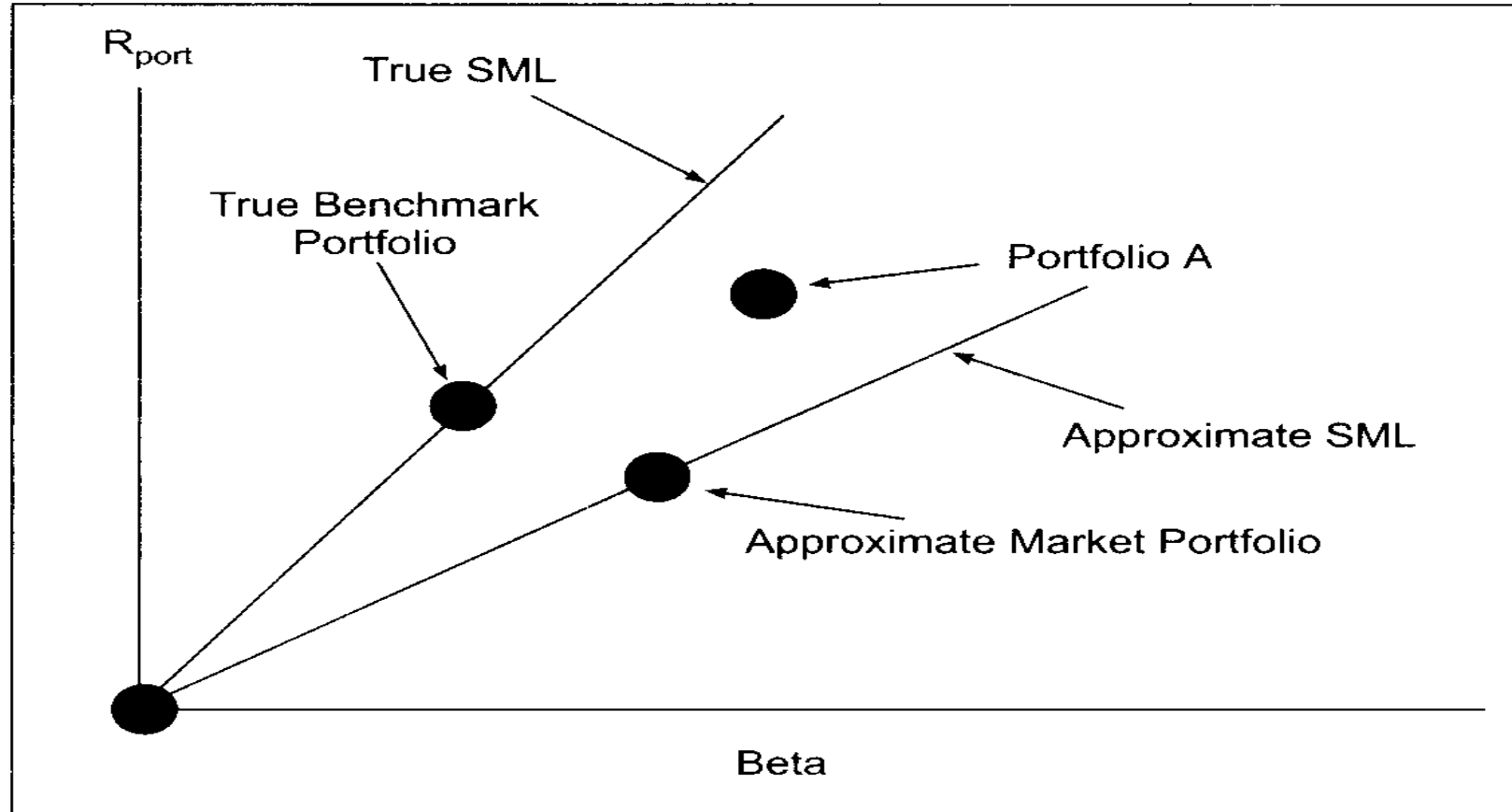


Index Portfolios

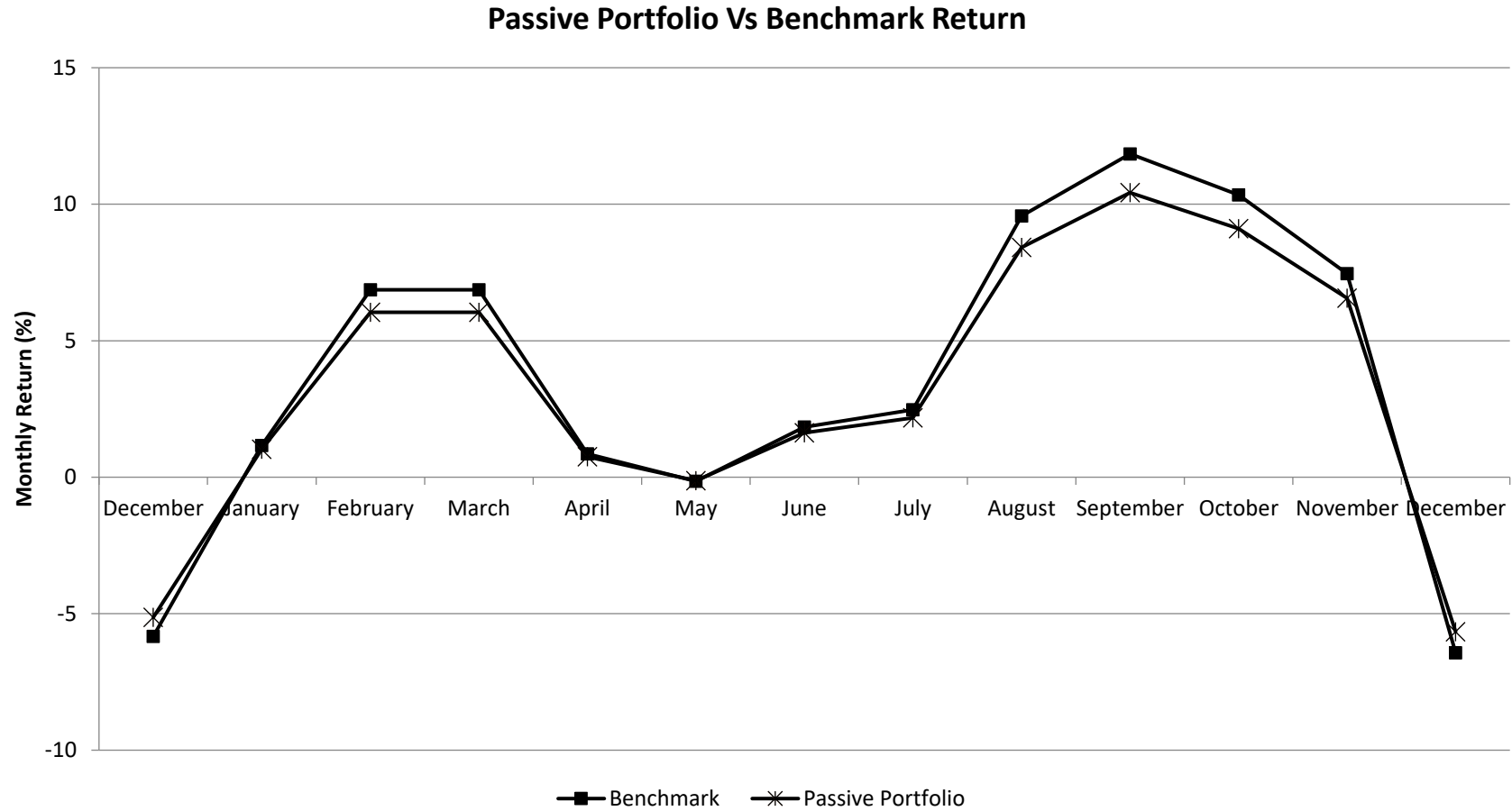
Benchmark Portfolios

- Benchmark portfolio represents the performance evaluation standard for portfolios before they are formed
- Benchmark portfolio are passive index portfolios
 - BSE Sensex, BSE 100, S & P Nifty
- Features of Benchmark Portfolios
 - Unambiguous
 - Investable
 - Measurable
 - Appropriate
 - Pre-specified

Benchmark Errors



Benchmark Portfolios



Index Portfolios

- Represent a financial market performance
- Indices are mathematical formulations that are constructed based on a group of representative instruments to measure value movements in markets such as shares, bonds and commodities
- Index Computation methods
 - Price average
 - Value weighted average

Price Average Method

- **Index** $I = \frac{\sum S_j}{D}$; $j = 1, \dots, N.$
- N = The number of stocks in the index
- S_j = Stock j market price; $j = 1, \dots, N.$
- D = Divisor
- Initially, $D = N$ and the Index is set at an agreed upon level.
 - To assure continuity, the Divisor is adjusted over time.

Value Weighted Average Method

- **Index**
$$I_t = \frac{\sum \omega_{tj} S_{tj}}{\sum \omega_{Bj} S_{Bj}} \quad j = 1, 2, \dots, N$$

- B = Predetermined Base Time Period
- Initially: $t = B$
- Initial value of the Index is set at an arbitrarily chosen value: M.

Index Adjustments

- Stock Split : Multiple shares are created
- Example: One share value of 40 is split into two implies value of split share is half of 40 i.e., 20.
 - Instead of one share value of 40 market now trades two shares with of value 20 each
- Before stock split $(S_1 + S_2 + \dots, S_N) / D_1 = I_1$
- Index $(60 + 40 + 50 + 30 + 20) / 5 = 40$ $I = 40$ and $D = 5$.
- After stock split $(S_1 + \frac{1}{2} S_2 + \dots, S_N) / D_2 = I_1$
- Index $(60 + 20 + 50 + 30 + 20) / D = 40$ $I = 40$ and $D = 4.5$

Index Adjustments

- Replacement of shares in the index
- Before replacement $(S_1 + S_2(A) + \dots + S_N)/D_1 = I_1$
- Index $(30 + 20 + 65 + 60 + 25)/5 = 40$ $I = 40$ and $D = 5$.
- After replacement $(S_1 + S_2(B) + \dots + S_N)/D_2 = I_1$
- Index $(30 + 100 + 65 + 60 + 25)/D = 40$ $I = 40$ and $D = 7$.

Index Adjustment

- Additional Share in the index
- Before share addition $(S_1 + S_2 + \dots + S_N) / D_1 = I_1$
- Index $(30 + 20 + 65 + 60 + 25) / 5 = 40$ $I = 40$ and $D = 5$.
- After share addition $(S_1 + S_2 + \dots + S_N + S_{N+1}) / D_2 = I_1$
- Index $(30 + 20 + 65 + 60 + 25 + 40) / 6 = 40$ $I = 40$ and $D = 6$.

Index Adjustment

- Dividend Distribution
- Before stock dividend $(S_1 + S_2 + \dots + S_N) / D_1 = I_1$
- Index $(60 + 40 + 50 + 30 + 20) / 5 = 40$ $I = 40$ and $D = 5$.
- Firm S_2 distributes 40% stock dividend
- Stock dividend reduces the price of the share by 40%
- After stock split $(S_1 + (0.6 \times S_2) + \dots + S_N) / D_2 = I_1$
- Index $(60 + 24 + 50 + 30 + 20) / D = 40$ $I = 40$ and $D = 4.6$.

BSE Indices

Equity

Market Cap /Broad

S&P BSE SENSEX
S&P BSE SENSEX 50
S&P BSE Bharat 22 Index
View all...

Sector & Industry

S&P BSE Commodities
S&P BSE Energy
S&P BSE Fast Moving Consumer Goods
View all...

Thematics

S&P BSE India Infrastructure
S&P BSE India Manufacturing Index
S&P BSE 500 Shariah
View all...

Strategy

S&P BSE Dividend Stability Index
S&P BSE SENSEX 2X Leverage Daily Index
S&P BSE SENSEX Futures Index
View all...

Sustainability

S&P BSE 100 ESG
S&P BSE CARBONEX

Volatility

S&P BSE SENSEX 1-Month Real Vol
S&P BSE SENSEX 2-Month Real Vol
S&P BSE SENSEX 3-Month Real Vol

Fixed Income

Composite

S&P BSE India Bond

Government

S&P BSE India Sovereign Bond Index
S&P BSE India Government Bill Index
S&P BSE India Provincial Bond Index
View all...

Corporate

S&P BSE India Financials Bond Index
S&P BSE India Services Bond Index
S&P BSE India Industrials Bond Index
View all...

Money Market

S&P BSE Liquid Rate Index

BSE Sectoral Indices

S&P BSE AUTO
S&P BSE BANKEX
S&P BSE CONSUMER DURABLES
S&P BSE CAPITAL GOODS
S&P BSE Fast Moving Consumer Goods
S&P BSE Telecommunication
S&P BSE Utilities
S&P BSE Commodities
S&P BSE Consumer Discretionary
S&P BSE Energy
S&P BSE Financial Services

S&P BSE Industrials
S&P BSE Healthcare
S&P BSE Information Technology
S&P BSE METAL
S&P BSE OIL & GAS
S&P BSE POWER
S&P BSE REALTY
S&P BSE TECK
S&P BSE Services

Index Constituents

- Representation of sectors in the economy
- Representative criteria: Liquidity, market capitalisation, and trading frequency.
- Companies are weighted based on the share of the total market capitalization in the index
- Divide the free float market capitalization of the 30 companies by the index divisor (index is comparable across changes in constituents / corporate actions)

Sensex Computation

- Free float market capitalization
- $(\text{Base value of Index} / \text{Base Free float Market Capitalization}) \times \text{Free float market capitalization of 30 companies constituting the index}$
- Free float market capitalization is the total market capitalization of the company available to the public for trading
- Free float: Total outstanding shares excluding shares held by promoters, governments, and other strategic investors that are not readily available for trading.
- Weighted average market capitalization method.

Sensex Computation

- Market capitalisation of each stock in the index multiplied by a predetermined weightage factor.
- Weightage factor is based on the free-float market capitalisation of the stock and its sectoral representation.
- Sum of the stocks' weighted market capitalisations is divided by a scaling factor to arrive at the Sensex value.

Sensex Constituents

SL	WEIGHT(%)	Name	Industry	Price	Mcap Full (Cr.)	M.Cap FF (Cr.)
1	12.95	HDFC Bank	Banks	1446.35	10,98,707.88	10,87,720.80
2	11.91	RIL	Integrated Oil & Gas	2958.1	20,01,358.50	10,00,679.25
3	9.09	ICICI Bank	Banks	1087.85	7,63,581.30	7,63,581.30
4	6.87	Infosys	IT Consulting & Software	1616.95	6,71,121.34	5,77,164.35
5	5.11	L&T	Construction & Engineering	3676.7	5,05,410.09	4,29,598.57
6	4.96	TCS	IT Consulting & Software	4110.1	14,87,070.15	4,16,379.64
7	4.36	ITC	Cigarettes,Tobacco Products	413.35	5,15,921.57	3,66,304.32
8	3.72	Axis Bank	Banks	1111.9	3,43,079.60	3,12,202.44
9	3.71	Bharti Airtel	Telecom Services	1203.5	6,77,448.44	3,11,626.28
10	3.6	SBI	Banks	788.15	7,03,393.29	3,02,459.11
11	3.06	Kotak Mahindra Bank	Banks	1749.35	3,47,513.86	2,57,160.25
12	2.57	HUL	Housing Finance	2420.65	5,68,753.81	2,16,126.45
13	2.18	TATAMOTORS	Passenger Cars & Utility Vehicles	1039.35	3,45,402.16	1,83,063.14
14	2.12	Bajaj Finance	Holding Companies	6419.3	3,96,727.55	1,78,527.40
15	2.06	Sun Pharma	Pharmaceuticals	1606.95	3,85,479.74	1,73,465.88

Sensex Constituents

SL	WEIGHT (%)	Name	Industry	Price	Mcap Full (Cr.)	M.Cap FF (Cr.)
16	2.06	HCL Tech	IT Consulting & Software	1637.05	4,44,240.54	1,73,253.81
17	1.99	M&M	Passenger Cars & Utility Vehicles	1896.25	2,35,804.15	1,67,420.95
18	1.99	NTPC	Electric Utilities	351.95	3,41,274.16	1,67,224.34
19	1.88	Titan Co	Other Apparels & Accessories	3780.65	3,36,249.01	1,58,037.03
20	1.81	Maruti Suzuki India	Passenger Cars & Utility Vehicles	11520	3,61,654.14	1,51,894.74
21	1.59	Power Grid	Electric Utilities	292.55	2,72,089.16	1,33,323.69
22	1.54	Tata Steel	Iron & Steel/Interm.Products	157.25	1,96,302.14	1,29,559.41
23	1.54	Asian Paints	Furniture,Furnishing,Paints	2870.1	2,75,299.36	1,29,390.70
24	1.33	UltraTech Cement	Cement & Cement Products	9672.6	2,79,236.04	1,11,694.42
25	1.23	IndusInd Bank	Banks	1562.2	1,21,722.91	1,03,464.47
26	1.09	Nestle India	Packaged Foods	2561.95	2,46,809.77	91,319.62
27	0.97	Tech Mahindra	IT Consulting & Software	1288.9	1,25,869.74	81,815.33
28	0.94	JSWSTEEL	Iron & Steel/Interm.Products	829.75	2,02,911.54	79,135.50
29	0.9	Bajaj Finserv	Finance (including NBFCs)	1582.6	2,52,502.06	75,750.62
30	0.87	Wipro	IT Consulting & Software	515.85	2,69,531.25	72,773.44

Sensex Index Computation

- Current Index: 73,833.22
- Prior Index: 73,667.9
- Points of change: 165.32
- $\text{Current Index} = \text{Prior Index} + \text{Points of Change}$
- Scaling Factor: 10.72
- Total weighted Index Market capitalization: 7,91,328.11 (Current)
- Total weighted Index Market capitalization: 7,89,556.24 (Prior)

Index Values

S&P BSE SENSEX		73667.96+165.32(+0.22%)		
Scrip Code	Company	LTP	Change Val	Points Contribution
500180	HDFCBANK	1459.9	32.85	218.38
532540	TCS	4191.35	69.85	62.69
500325	RELIANCE	2950.2	19	57.43
500209	INFY	1612.2	12.75	39.87
532454	BHARTIARTL	1204.4	7.3	16.18
532500	MARUTI	11506.75	113.65	13.24
500520	M&M	1893.5	2.9	2.43
532755	TECHM	1292	2.35	1.29
500114	TITAN	3749.8	-0.25	-0.09
500820	ASIANPAINT	2875.55	-1.6	-0.63
532898	POWERGRID	285.05	-0.45	-1.79
532978	BAJAJFINSV	1593.1	-6.8	-3.23
532281	HCLTECH	1635	-3.6	-3.34
532187	INDUSINDBK	1532.2	-8	-4.57
507685	WIPRO	510.65	-3.75	-4.63

Continued.....

Index Values

Scrip Code	Company	LTP	Change Val	Points Contribution
500034	BAJFINANCE	6434.1	-22.1	-5.38
532174	ICICIBANK	1076.95	-1	-6.14
500790	NESTLEIND	2589.25	-23.15	-7.23
500470	TATASTEEL	152.45	-1.05	-7.42
524715	SUNPHARMA	1584.4	-8	-7.56
500696	HINDUNILVR	2378.15	-9.95	-7.78
532538	ULTRACEMCO	9612.75	-85.15	-8.61
500228	JSWSTEEL	817.2	-12.2	-10.19
500247	KOTAKBANK	1720.35	-8.85	-11.09
532555	NTPC	344.85	-2.95	-12.28
500570	TATAMOTORS	1016.6	-11.4	-17.92
500510	LT	3621.1	-17.5	-18.53
532215	AXISBANK	1096.25	-8.2	-20.37
500875	ITC	404.25	-5.15	-39.94
500112	SBIN	759.4	-14.1	-47.4
Total				165.39

Points Contribution

Code	Firm	WEIGHT (%)	M.Cap FF (Cr.)	Mcap Full (Cr.)	Index Representation	Prior Index Representation	Points Contribution
500180	HDFCBANK	12.95	10,87,720.80	10,98,707.88	142282.6705	139942.122	218.38
532540	TCS	4.96	4,16,379.64	14,87,070.15	73758.67944	73086.7819	62.69
500325	RELIANCE	11.91	10,00,679.25	20,01,358.50	238361.7974	237746.2753	57.43
500209	INFY	6.87	5,77,164.35	6,71,121.34	46106.03606	45678.71823	39.87
532454	BHARTIARTL	3.71	3,11,626.28	6,77,448.44	25133.33712	24959.92347	16.18
532500	MARUTI	1.81	1,51,894.74	3,61,654.14	6545.939934	6404.036545	13.24
500520	M&M	1.99	1,67,420.95	2,35,804.15	4692.502585	4666.458383	2.43
532755	TECHM	0.97	81,815.33	1,25,869.74	1220.936478	1207.110544	1.29
500114	TITAN	1.88	1,58,037.03	3,36,249.01	6321.481388	6322.445988	-0.09
500820	ASIANPAINT	1.54	1,29,390.70	2,75,299.36	4239.610144	4246.362345	-0.63
532898	POWERGRID	1.59	1,33,323.69	2,72,089.16	4326.217644	4345.402468	-1.79
532978	BAJAJFINSV	0.9	75,750.62	2,52,502.06	2272.51854	2307.136965	-3.23
532281	HCLTECH	2.06	1,73,253.81	4,44,240.54	9151.355124	9187.152505	-3.34
532187	INDUSINDBK	1.23	1,03,464.47	1,21,722.91	1497.191793	1546.172041	-4.57
		100.00	84,02,117.25		7,91,328.11	789556.2424	165.39

Weight: (M Cap FF / Total)%

Index Representation: Weight x Mcap Full

Continued.....

Points Contribution

Code	Firm	WEIGHT (%)	M.Cap FF (Cr.)	Mcap Full (Cr.)	Index Representation	Prior Index Representation	Pts Contribution
507685	WIPRO	0.87	72,773.44	2,69,531.25	2344.921875	2394.54519	-4.63
500034	BAJFINANCE	2.12	1,78,527.40	3,96,727.55	8410.62406	8468.285709	-5.38
532174	ICICIBANK	9.09	7,63,581.30	7,63,581.30	69409.54017	69475.34733	-6.14
500790	NESTLEIND	1.09	91,319.62	2,46,809.77	2690.226493	2767.716032	-7.23
500470	TATASTEEL	1.54	1,29,559.41	1,96,302.14	3023.052956	3102.578873	-7.42
524715	SUNPHARMA	2.06	1,73,465.88	3,85,479.74	7940.882644	8021.90905	-7.56
500696	HINDUNILVR	2.57	2,16,126.45	5,68,753.81	14616.97292	14700.35723	-7.78
532538	ULTRACEMCO	1.33	1,11,694.42	2,79,236.04	3713.839332	3806.119406	-8.61
500228	JSWSTEEL	0.94	79,135.50	2,02,911.54	1907.368476	2016.58264	-10.19
500247	KOTAKBANK	3.06	2,57,160.25	3,47,513.86	10633.92412	10752.78428	-11.09
532555	NTPC	1.99	1,67,224.34	3,41,274.16	6791.355784	6922.970105	-12.28
500570	TATAMOTORS	2.18	1,83,063.14	3,45,402.16	7529.767088	7721.829681	-17.92
500510	LT	5.11	4,29,598.57	5,05,410.09	25826.4556	26025.05604	-18.53
532215	AXISBANK	3.72	3,12,202.44	3,43,079.60	12762.56112	12980.88227	-20.37
500875	ITC	4.36	3,66,304.32	5,15,921.57	22494.18045	22922.24853	-39.94
500112	SBIN	3.6	3,02,459.11	7,03,393.29	25322.15844	25830.18115	-47.4
		100.00	84,02,117.25		7,91,328.11	789556.2424	165.39

National Stock Exchange Indices

- S&P CNX nifty
- CNX nifty junior
- CNX 100
- S&P CNX 500
- CNX midcap
- Nifty midcap 50
- S&P CNX defty
- Energy index
- Pharma index
- Service sector index
- MNC index
- PSE index
- FMCG index
- IT index
- Bank index